

Relation between 1-norm of matrix and Euclidean norm of vectors

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Suppose I have column vectors u, v in a Hilbert space. I can define rank-1 matrices uu^* and vv^* , where $*$ denotes the transpose conjugate. If it is the case that

0

$$\|uu^* - vv^*\|_1 \leq \varepsilon,$$



where $\|\cdot\|_1$ is the trace norm or nuclear norm or Schatten 1-norm defined as $\|A\|_1 = \text{Tr}(\sqrt{A^*A})$, can one say something about an upper bound for



$$\|u - v\|?$$



Here $\|\cdot\|$ is the Euclidean 2-norm for vectors.

In my case, I have the additional knowledge that uu^* and vv^* are positive semidefinite with unit trace if that helps with the proof.

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asked Nov 28, 2020 at 18:37

 user1936752
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1 Answer

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The most you can say is $\|u - v\| \leq 2$.

1

First of all, notice that $\text{Tr}(uu^*) = 1 \implies \|u\| = 1$, and the same holds for v . As a consequence,



$$\|u - v\| \leq \|u\| + \|v\| = 2.$$

Take now $v = -u$. We have that



$$vv^* = uu^* \implies \|vv^* - uu^*\| = 0 \leq \varepsilon$$

for any $\varepsilon > 0$, but



$$\|u - v\| = \|2u\| = 2.$$

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edited Nov 28, 2020 at 19:58

answered Nov 28, 2020 at 19:11

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